Medical Physics Department

Physiology Lab.

2nd Class

2025-2026

Lab 1

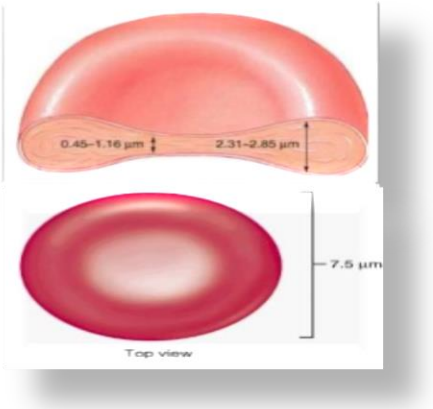
Total Red Blood Cell Count



By: Nadia Khalid Mustafa

Introduction

Red blood cells (RBCs) are the round shape, biconcave discs, present in the blood that helps in the transport of gases throughout the body. The biconcave shape helps the RBCs in rendering the red cells quite flexible so that they can easily pass through the capillaries. On an average, the size of the Red Blood Cells (RBCs) is 7.2 – 7.4 mm (microns)



Functions of the RBCs:

1. Transport O2 from the lungs to the various tissues of the body and CO2 from the tissues to the lungs and this function is related to the Hb in the RBC
2. Regulation of blood pH which is due to the presence of carbonic anhydrase in the RBC

Disorders of Erythrocytes:

An imbalance between the rate of RBC production (erythropoiesis) and the rate of destruction.

• There are mainly two types:

 Polycythemia: excess of RBCs production resulting in increased viscosity of the blood, reduced blood flow rate, and, if sever, plugging of the capillaries.

 Anemia: deficiency in either RBCs or hemoglobin or both of them. Its types Nutritional anemia (most common).

Red Blood Cell Testing Methods

1. **Manual Method** (Hemocytometer): the **Neubauer chamber** remains the most common method used for cell counting around the world

2. **Automated (Electronic Cell Counting) Method :** This technique uses changes in electrical resistance to count cells and provide an assessment of cell volume . It is used to measure RBC in blood samples as well as samples of body cavity fluids .
3. **Microscopic examination**:manual review of blood smear to assess cell morphology and identify abnormalities

Manual Method (Hemocytometer Method)

Objective (Aim) of The Experiment: To determine the total number of RBCs evaluate the general shape of the RBCs for possible presence of certain disorders ,To determine the presence of RBC disorders (polycythemia or anemia)

• Diluted blood sample is loaded into a counting chamber (hemocytometer).

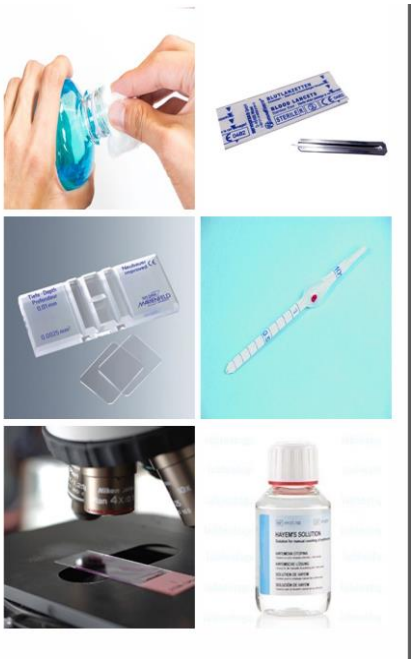
• RBCs are distributed evenly in a known volume.

• Cells are counted under a microscope and the number per microliter is calculated.

◇ **Basic Requirements:**

- 70% alcohol and cotton
- Sterile blood lancet
- Special RBC pipette
- Special slide (hemocytometer)
- Cover slide
- Special dilution fluid (Hayem′s Solution)

  or (Gower′s Solution)
- Microscope



Procedure:

1- RBC Pipette



- Graduated (0.5 IU – 101 IU)
- Large mixing chamber
- Used to withdraw blood sample, and then dilute it with (Hayem's or Gower's solution)

2- Dilution Fluid:

- **Hayem's Solution:**

This type of solution <u>can not</u> destroy WBCs and Platelets



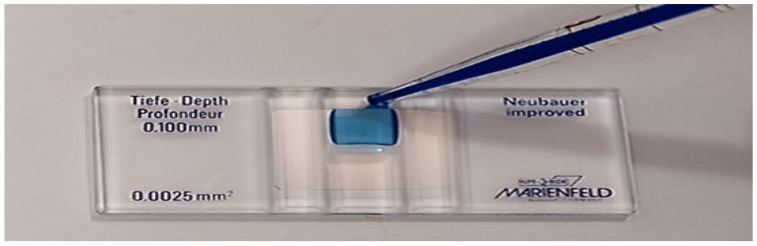
- **Gower's Solution:** This type of solution <u>can</u> destroy WBCs and Platelets

3- Hemocytometer (Counting Chamber)

Hemocytometer is a special slide containing a counting chamber, It is used for manual measurement of different types of cells in blood and other body fluids

• Used to determine the total number of:

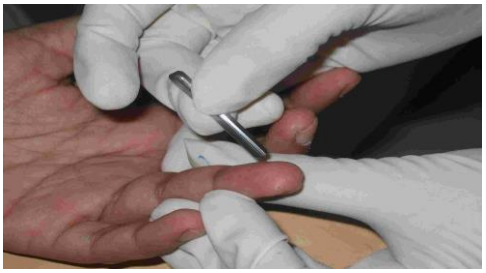
- WBCs in a blood sample, cerebrospinal fluid sample, and synovial sample
- RBCs in a blood sample
- Platelets in a blood sample



Procedures

1- First clean the Neubauer chamber and coverslip with cotton deep it in alcohol, then fixed coverslip on chamber.

2- Wipe the finger with cotton soaked with alcohol and allow it to dry. With a sterile disposable lancet do small prick on the fingertip and discard



the first drop because it contain tissue fluid.

3- When a drop of reasonable size has collected, hold the red blood cell pipette slightly tilted from the vertical position, apply its tip to the drop and aspirate blood to the mark 0.5.

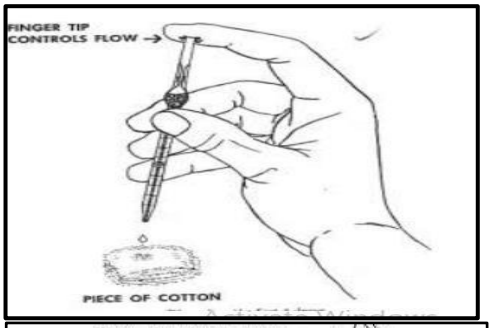


4- Draw the diluting Hayem's fluid up to the 101 mark, thus making 1:200 dilution of blood.

5- Clean the outside of pipette with cotton.

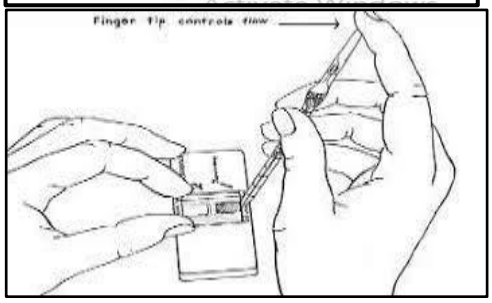
6- Mix continuously for 2-3 minutes.

7- Before Load the chamber by diluiting blood, discard the first four to five drops of the mixture on apiece of cotton to expel the diluent from the stem.



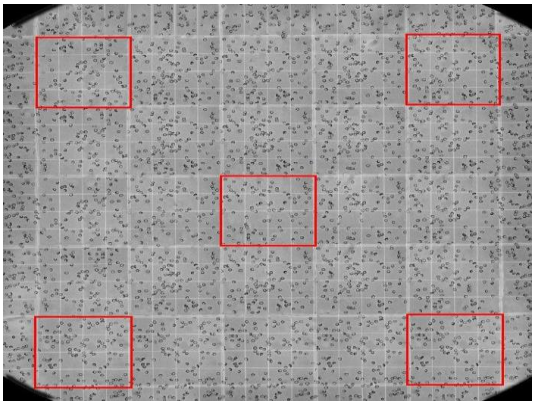
8- Load the chamber by diluting blood, let the drop fall near the cover glass and the chamber will fill due to capillarity. The excess fluid will flow in the moats.

9- Allow 2 min. for setting of the cells and then count.



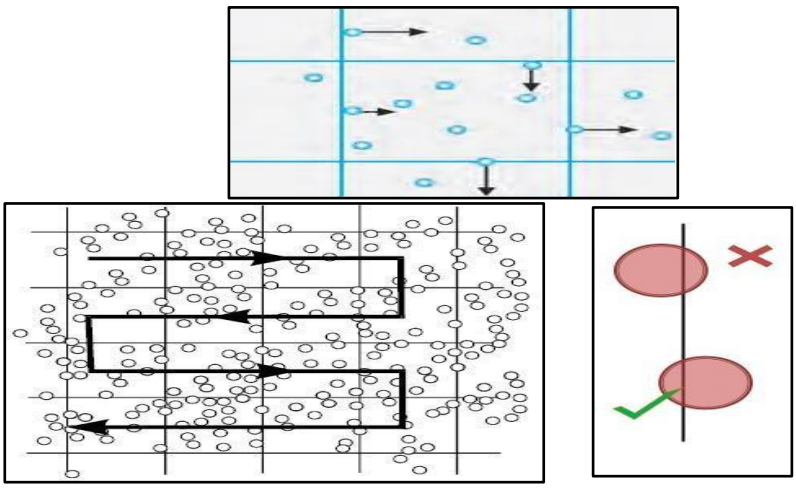
10- Place the chamber on stage of microscope.

11- Scan by using the low power (10X) objective lens. The cells should be distributed evenly in all of the squares.



12- Then count the RBC using 40x in Central Square. The central square is subdivided into 25 squares. Out of 25 squares, count four at each corner and one at center.

13- Counting rule, cells that touch the top and left lines should be counted while cells that touch the bottom and right lines should be ignored to avoid double counting with the cells in center of square.



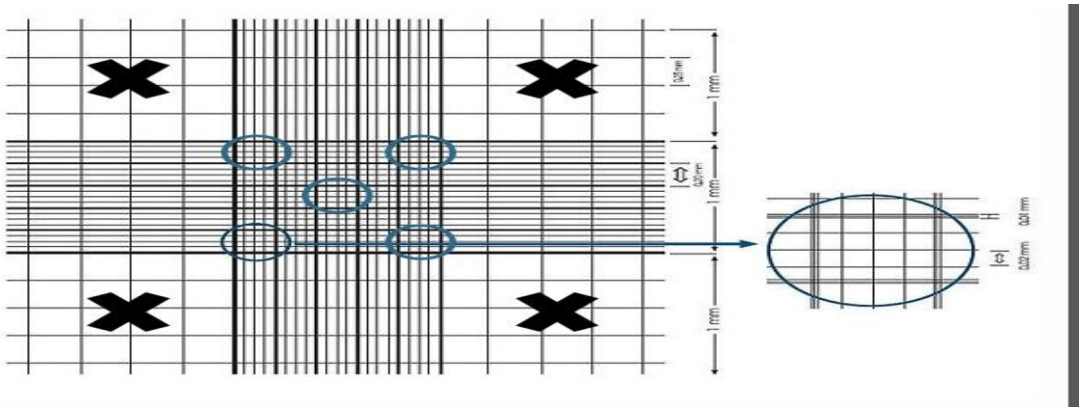
Calculation

Calculation Count the number (N) of cells in 80 small squares located in 5 middle-sizes squares (four located at the four corners and one in the middle). The size of 80 small squares in which “N” numbers of cells are found is:

1/20 x 1/20 x 1/10 x 80 = 1/50 mm3 .

1/20 mm is the sideline of the square, 1/10 mm is the depth of the counting chamber between cover slip and the ruling, 80 is the number of small squares used to count.

Therefore the total numbers of cells in 1 mm3 are = N x 50 (diluted sample) The actual total number of cells before dilution should be: N x 50 x 200 = N x 1000



Example:

• Determine the RBC counting for a male patient if you have the following readings: (R1= 110, R2= 135, R3= 98, R4= 120, R4=84). State if the result is normal, Anemia or Polycythemia?

Answer:

R total = R1 + R2 + R3 + R4 + R5 = 110 + 135 + 98 + 120 + 84 = 547

Since $1mm^3$ (B) = 10000R

Then 10000 * 547 = 5470000 RBCs / $1mm^3$ of blood

Normal range:

Male: 4.6- 6 million	Female: 4.2- 5 million
Children: 4.6- 5.5 million	Newborn: 6 - 7.2 million

Factors Influencing Red Blood Cell Count

1. **Genetic Factors:** Inherited conditions and genetic variations can significantly impact red blood cell production, shape, and lifespan, including conditions like sickle cell disease.
2. **Environmental Conditions** High altitude living stimulates increased production due to lower oxygen levels. Climate and air quality also play important roles.
3. **Nutritional Status** Iron, vitamin B12, folate, and protein intake directly affect red blood cell production and quality. Deficiencies can lead to anemia.
4. **Physical Activity** Regular exercise can increase red blood cell count, whilst extreme endurance training may temporarily decrease levels due to increased plasma volume.